

# **DashAR: An Implementation of Augmented Reality Technology for Transportation Applications**

A Thesis Defense Presentation by Trevor D. Brown

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## Presentation Flow

01

### History

XR, Automobile Instrumentation, and the Heads Up Display.

02

### Opportunity

A new approach to Automotive HUDs.

03

### Previous Research

Prior investigations into Automotive HUDs.

04

### Thesis Objectives

What can become the modern Automotive HUD?

05

### Timeline

Planning for the future.

06

### Design

Designing an AR-based Automotive HUD System.

## Presentation Flow (continued)

07

### Implementation

Developing the DashAR System.

08

### End Results

DashAR in Action.

09

### Hurdles and Problems

Development dilemmas and the race against time.

10

### Future Works

What's next for DashAR?  
Where can it go next?

11

### Conclusions

Let's wrap this up.

12

### Thanks

Expression of my gratitude.

01

# History

Automotive Instrumentation, Extended Reality (XR), and the Heads Up Display (HUD)

## History of Extended Reality (XR)



### Stereopsis

1838, by Charles Wheatstone

## History of Automotive Instrumentation



### **Ammeter**

(1925)  
Ford Model T

## History of Automotive Instrumentation (cont.)



### **Various Gauges**

(2017)  
Subaru Forester

# History of Automotive Instrumentation (continued)



## Various Gauges

(2017)  
Subaru Forester

# History of XR (continued)



## Sensorama

1956, by Morton Heilig

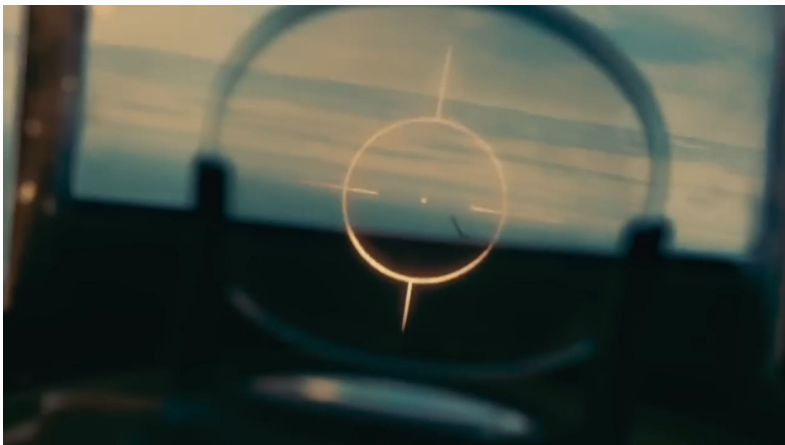
## History of XR (continued)



### **The Sword of Damocles**

1968, by Ivan Sutherland

## History of Heads Up Displays (HUDs)



### **Fighter Plane HUDs**

(1942 - present)

Frame from *Dunkirk* (2017)

## History of HUDs (continued)



### Automobile HUDs

(1980 - present)

BMW E60

02

# Opportunity

A new approach to Automotive HUDs.





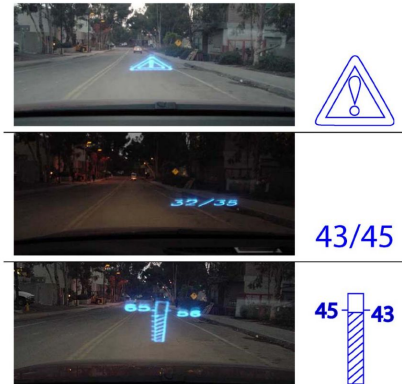
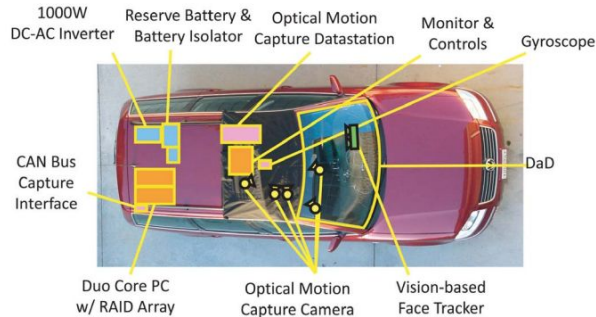
03

# Research

Prior investigations into Automotive HUDs.



## Literature Review



A. Doshi, S. Yuanhsien, et. al. (2009). *A Novel Active Heads-Up Display for Driver Assistance*. IEEE Transactions on Systems, Man, and Cybernetics - Part B.

- Windshield Implementation of a HUD while driving; alerts of adverse conditions.

## Literature Review (continued)

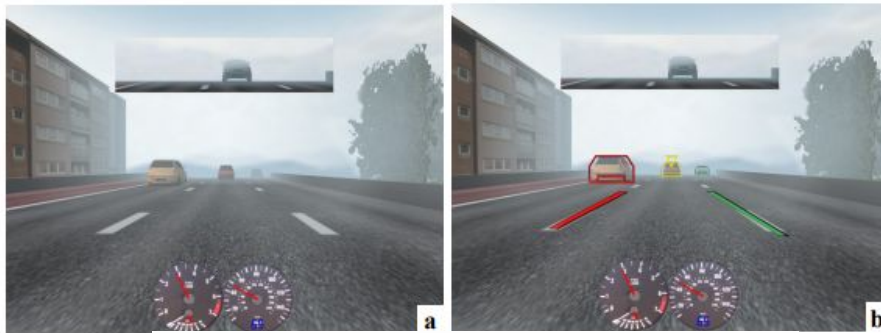


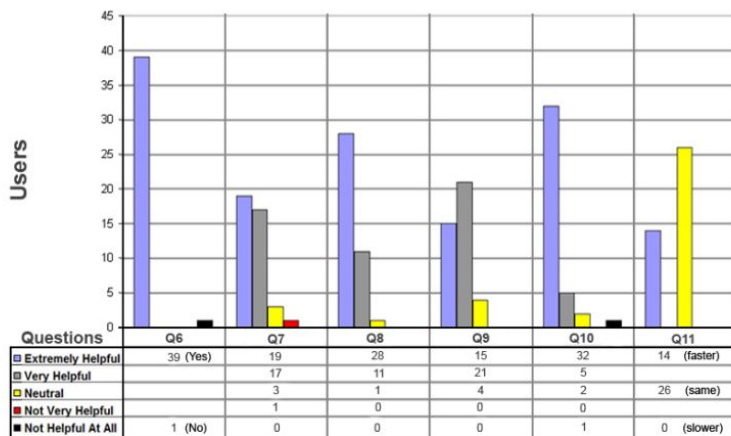
Figure 1: Driving under low visibility conditions  
a) With contemporary HDD, and b) With proposed HUD visual aid

V. Charissis and S. Papanastasiou. (2006). *Human-Machine Collaboration Through Vehicle Head Up Display Interface*. European Annual Conference on Human Decision-Making and Manual Control.

- Heads Up vs. Heads Down Display. Consideration for low visibility conditions.

# Literature Review (continued)

Chart 1. Evaluation outcomes of comparison between HUD and HDD.



## Questions of Interest

Q6: Would you use this HUD interface navigation system under low visibility conditions?

Q7: How easy was [it] for you to get used to the HUD interface?

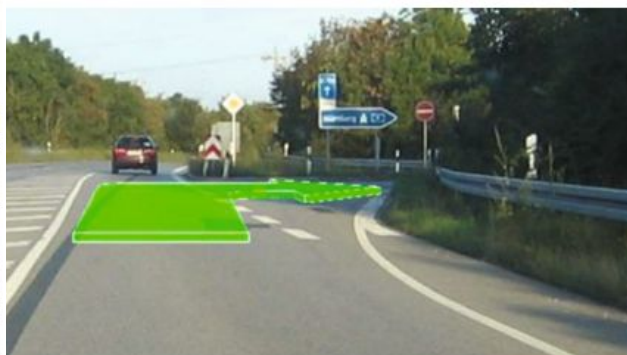
Q8: Do you think it would be a helpful system to integrate in future vehicles?

Q9: How helpful was the HUD for decreasing your stress level during driving in low visibility?

Q10: How helpful was the HUD for minimising your stress level compared to HDD?

Q11: When using the HUD, were you driving faster, the same, or slower?

# Literature Review (continued)

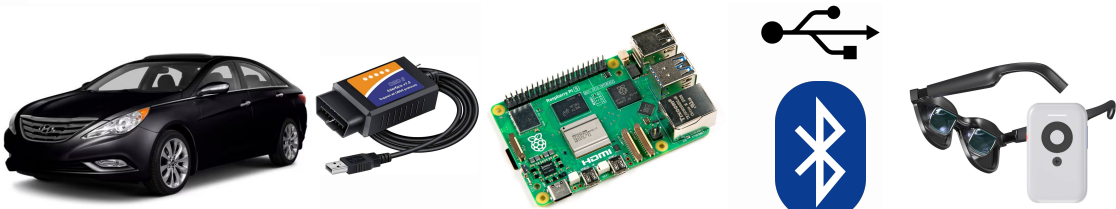
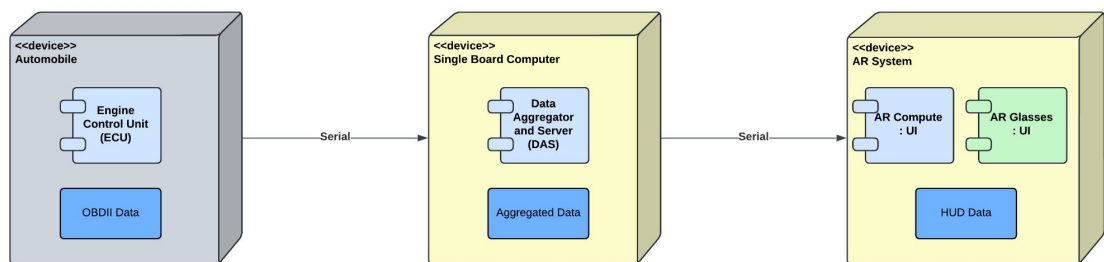


- A. Pauzie. (2015). *Head Up Display in Automotive: A New Reality for the Driver*. International Conference on Human-Computer Interaction.
  - Reasoning for improved "eyes on the road" time using HUD.

04

# Objectives

Building a AR-based HUD Framework.



## Methodology



**Agile**

### **Speed**

Faster, but unpolished results.  
Gradually refined.

### **Efficiency**

Don't Repeat Yourself, Keep It Simple  
\*Silly\* (DRY KISS)

### **Simplicity**

Concise, well-documented interfaces.

### **Modularity**

Stable, extensible HUD framework and  
API.

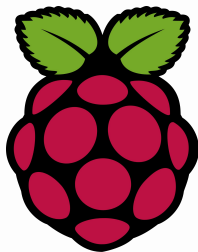
## Target Platforms



## Target Platforms (continued)



## Development Tools



## Project Schedule (Planned)

[illegible]

## Project Schedule (Actual)

[illegible]

# Thesis Artifacts

## Public and Research Community

- PoC software system, useful in automobiles
- Source Code
- System Implementation and Utilization Documentation
- Test Cases with Results
- Peer Reviewed Publication(s)
- Conference Presentation(s)

## The Graduate School at WKU

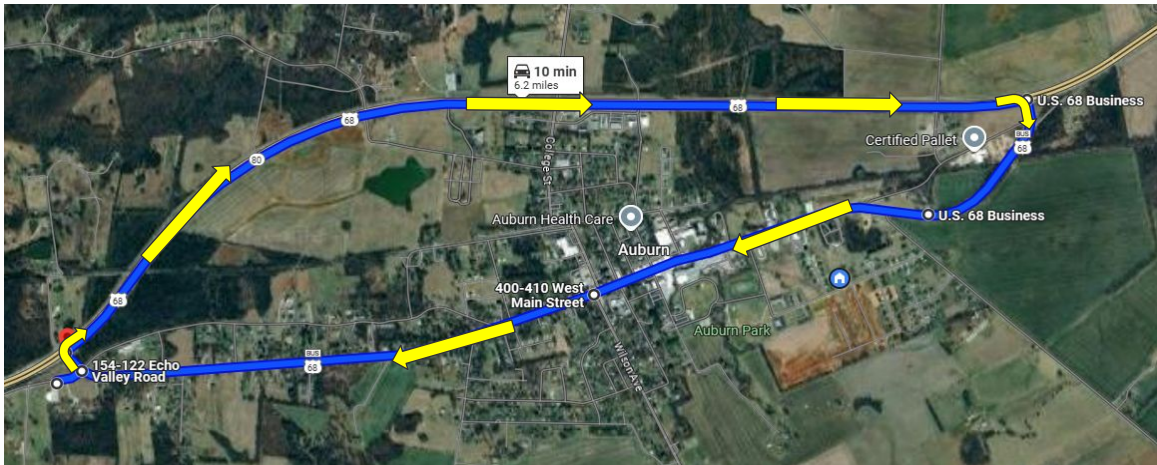
- Submission of Thesis
- Submission to TopSCHOLAR

# Demonstration

Using an AR-based HUD.



## Test Drive (11/10/2024)



## Technical Challenges (thus far...)

### Data Aggregator and Server

Safe power down (e.g. UPS, battery pack), establishing third-party API connectivity (e.g. cellular modem, hotspot)

### HUD

Data request methodology (e.g. Client/Server (synchronous), Pub/Sub (asynchronous), etc.), non-distracting UI, dynamic UI through configurations, Restrictions to additional apps while driving.

### UX

Intuitive and non-distracting, usability in any conditions (rain, snow, night, etc.)

## Approximate Cost

### Data Aggregator and Server

Raspberry Pi 5 (Computer and Power Supply) - \$92

12V to 120V Power Inverter (150W) - \$40

Surge Protector - \$10

### HUD

XREAL Air 2 Pro - \$400

XREAL Beam Pro - \$200

### Total Cost

\$ 810

**05**

# Future Works

Where can this research go next?

## Planes, Trains, and Automobiles!

### Operational Data

Instrumentation, Operating Conditions, Weather, Hauled Load Data (for trucks), Customized Navigation (for military applications)

### Intelligent Operation

Computer Vision applications (Blind Spot Detection, Object Detection), Operating Limit Monitoring and Alerts (e.g. unsafe vehicle acceleration, excess load weight when hauling, etc.)

### Luxuries

Infotainment Integration, Climate Control, Multi-view Perspectives via camera feeds.

# Conclusions

Let's Begin!

## Conclusions

**Automotive HUDs  
are not fully  
realized.**



Restrictive implementations,  
limited information.

**Let's make a better  
Automotive HUD.**



Utilize new AR technology to  
solve a problem centered in  
distracted driving.

**Aim for Acceptance  
through Open  
Source.**



Publish findings, provide  
reproducible, cost effective  
solution usable in any modern  
car, using any AR headset or  
data collection methods. An  
Open Source framework,  
extensible by free and non-free  
modules.

# Do you have any questions?

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# Thanks

Expression of my gratitude.

## Family and Friends

My Lord and Savior, Jesus Christ

Jillian Brown (my wife!)

Jason Brown (my father!)

Dee Dee Brown (my mother!)

Ethan and Rhonda Brown (my brother and sister-in-law!)

Pastor Luke Brown (my brother!)

Mark Jeffcoat (my father-in-law!)

Kristin Jeffcoat (my mother-in-law!)

Meg Jeffcoat (my sister-in-law!)

Garland Brown (my grandfather!)

Betty Brown (my grandmother!)

Nan Stratton (my late grandmother!)

Brandon Brown (my uncle!)

Jennifer Brown (my aunt!)

Lydia Brown (my cousin!)

Matthew Brown (my cousin!)

The Browns, Brownings, Strattons, Smiths, Croppers, and Marshalls (my extended family!)

## Family and Friends (continued)

Seth, Christian, Barrett, and Noah Stratton

Cody, Delsie, and Olivia Meacham

Nathan Duncan

Michael Bailey

Liam Seymour

Christopher Goulet

Chandler Staggs

Travis Anderson

Katie Tyson

My classmates at WKU.

Dylan, Christina, and Lana Moore

DJ, Mary, and Cooper Lee

Ben, Erin, and Reuben Russell

Christopher and Jackie Bridgman

Casey Fortney

Alex, Tanya, and Ezra Puleo

Blake, Kristin, and Avery Monarch

Zach and Brooke Oak

Michael and Sabrina Newton

Alex, Nicole, and Violet Taylor

## Family and Friends (continued)

Kevin Lunsford

The Galloway Family

Tiny (posthumously) and Brenda Reagan

Nancy Scott

Greg Kunkel

Amy James

Judy Mullendore

Lisa Shipley (posthumously)

Mary Nunn

Brandi Fowler

Eric Wolfe

Heather Stubblefield

Kelly Jo Eysenbach

Barbara Scheidt

Al Smith

John Gregory

The staff and student workers of WKU ITS.

My colleagues at Logan Aluminum.

So many more that I can't list!

## Mentors and Educators

Melissa Jones (posthumously)

Samuel Kessler

John Jukes

Tommy Sandlin

Tyler Davenport

Dr. Michael Galloway

Dr. Rong Yang

Dr. Qi Li

Dr. Guangming Xing

Dr. James Gary

Dr. Mustafa Atici

Dr. Huanjing Wang

Dr. Zhonghang Xia

Dr. Mark Cambron

Dr. Stacy Wilson

Dr. Greg Arbuckle

The Faculty and Staff of the Logan County KY Board of Education.

The Faculty and Staff of WKU's School of Engineering and Applied Sciences.

## Mentors and Educators (continued)

William Wells

Rob VanFleet

Ricardo Ortolani

Nick Schoenbaechler

Aaron Harris

Brent Haselhoff

Brandon Vincent

Tyler Vincent

Amy Wirth

Dave Craig

David Brooks

Mike Hoots

Robert Malcomson

Julie Cox

Miranda Logsdon

Amy Mooneyhan

Harvey Harris

Jim Kelly

Marisa Hopkins

Sharon Dockins

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- Node.js (<https://nodejs.org/en>)
- Python (<https://www.python.org/>)
- OBDII  
(<https://learn.sparkfun.com/tutorials/obd-ii-uart-hookup-guide/obd-commands>)

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  - Sensorama
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  - The Sword of Damocles
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  - Automotive HUD
    - [https://en.wikipedia.org/wiki/Head-up\\_display](https://en.wikipedia.org/wiki/Head-up_display)